



# 3alemni Educational Platform



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## Abstract

The 3alemni platform aims to enhance education accessibility and structure by integrating intelligent tools such as facial recognition for attendance, geolocation-based check-ins, AI-generated flashcards and reels, and an assistant hiring system. It supports students in course enrollment and material access while providing teachers with robust dashboards and assistant hiring tools. The platform stands out by combining academic management, AI tutoring, and real-time interaction in one centralized system.

## Methods and Materials

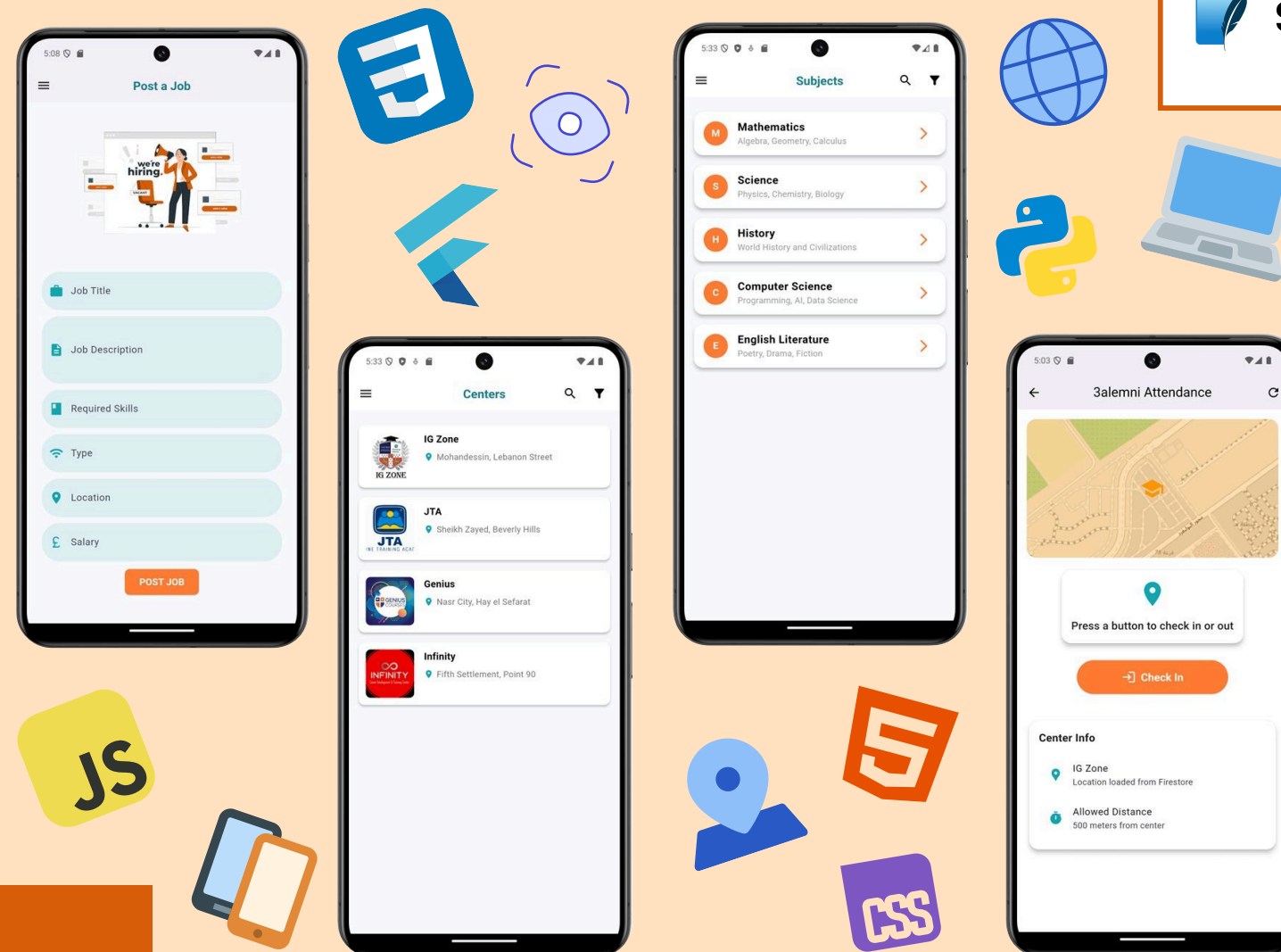
Frontend : Mobile: Flutter Web: HTML, CSS & JavaScript  
Backend : Flask + Python  
Databases : Firebase (Mobile) SQLite (Web)  
AI Components : Flashcards (Bart)  
Video Reels (Gemini + Google TTS)  
Face Recognition (FaceNet)  
GPS + Geofencing for Location Check-in  
Course Recommendation Chatbot

## Technologies & Tools Used

HTML Python  
CSS Natural Language Processing  
Javascript Facenet  
Flutter Text-Speech  
SQLite Named Entity Recognition

## Introduction

Students often struggle with fragmented educational tools, inefficient tutor discovery, and unfair assistant hiring practices. Inspired by personal academic struggles, we developed 3alemni – an integrated educational platform that connects students, teachers, and assistants through AI-driven features. Our goal was to eliminate the inefficiencies of traditional tutoring ecosystems and enhance education using smart automation, performance analytics, and structured interaction.



## Discussion

3alemni bridges several gaps in the Egyptian educational system by offering a digital solution for real-time learning, teaching, and academic support. The facial recognition and geolocation-based attendance tools proved highly effective in hybrid and online learning contexts. Unlike many global LMS tools, our solution focuses on the local market (Thanawya Amma, IGCSE, American diploma, college) while incorporating personalized AI-driven support, offering an edge over existing platforms.

## conclusions

3alemni successfully integrates AI into everyday academic workflows. Its modular design and high usability scores make it scalable for future deployment in schools and educational centers. With reliable testing results and user satisfaction, it is a strong candidate for educational transformation in Egypt. Planned improvements include adding parent accounts, enhancing chatbot voice interaction, and progress tracking using visualization.

| Feature / Platform       | Technology / Method                  | Accuracy      |
|--------------------------|--------------------------------------|---------------|
| Facial Recognition       | FaceNet (InceptionResnetV1 + MTCNN)  | 98.21%        |
| Geolocation (Website)    | Browser Geolocation API              | 30–100 meters |
| Geolocation (Mobile App) | Flutter Geolocator Plugin            | 3–10 meters   |
| AI Flashcards            | Google Bart / Gemini                 | 95%           |
| AI Video Reels           | Gemini (Text-to-Video Summarization) | 90%           |